A positive residual in the quadratic memory sector
Fixed-charge norm and the completed equal-time Gram

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Abstract

We evaluate the missing half-filled norm of the derivative singlet. A Casimir identity for the exchanged contraction reduces the adjacent-charge calculation to two known Racah kernels. An explicit expansion cancels the terms of order p^2 and p; logarithmically growing windows control the remainders and the complete input-rank sum. Combined with the preceding cross, this proves a positive output-deficit residual after projection onto the three existing directions H^2, A, and $J_H F$.

1 Setting and main result

Use standard CAR, the full overlapping complex Gaussian p-body ensemble, $\hbar = 1$, and the normalized maximally mixed trace at $m = N/2$. Inner products include the disorder average. Let $h = H_m$, $k = H_{m-1}$ use the same couplings and set

$$C = \mathbb{E}\,\mathrm{tr}_m\, h^2, \quad B = \mathsf{W}_2 h^2, \quad O = \mathsf{W}_2 A, \quad A = \mathrm{Rem}(k^2) + (2p - m)h^2. \tag{1}$$

W_2 is ordinary Gaussian second chaos. The output-deficit operator is $\mathcal{S}_N = \sum_{a\le b,r}(a+b-r)\mathsf{W}_2 Z^+_{ab;r}$, with the rank-product convention of the preceding notes. First take $p^2/N \to \lambda > 0$, $C \to J_s^2/\lambda$; write $q = e^{-4\lambda}$ and $\theta = 2\lambda$. We prove

$$\frac{\|O\|^2_{2,m}}{C^2} \longrightarrow 4\lambda + 4\lambda^2(1+q), \tag{2}$$

$$\|O - \mathcal{S}_N\|^2_{2,m} \longrightarrow \Delta_* := \frac{2J_s^4(1+q)}{\lambda} > 0. \tag{3}$$

Let $V_S = J_s^4[2(1-q)/\lambda + 16q/(1+q)]$ be the previous output-deficit variance after the B subtraction. Projection onto the span of $B, O, J_H F$ leaves

$$V_{\mathrm{new}} = \frac{V_S \Delta_*}{V_S + \Delta_*} = \frac{J_s^4(1+q)(1-q^2+8\lambda q)}{\lambda(1+q+4\lambda q)} > 0. \tag{4}$$

After this double-scaled limit, $\lambda \to 0^+$ gives $V_{\mathrm{new}} \to 16J_s^4$. The result concerns this specified equal-time operator span in the fixed half-filled state.

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2 The exchanged Casimir insertion

This step works at any charge M away from the trivial endpoints. For independent isotropic Hermitian rank-a, b tensors U, V with $\mathbb{E}U^2 = \mathbb{E}V^2 = I$, define

$$\mathcal{C}X = \frac{1}{2}\sum_{ij}[E_{ij}, [E_{ji}, X]], \qquad \mathcal{C}U = x_a U, \quad x_a = a(N+1-a). \tag{5}$$

Write $\Phi_a(X) = \mathbb{E}UXU$, whose rank-$b$ eigenvalue is the known neutral Racah kernel $\chi_{ab}^{(M)}$. The ordered-product moment from the earlier covariance calculation and the exchanged-product moment are

$$\begin{aligned} \mathbb{E}\,\mathrm{tr}_M (UV)^\dagger \mathcal{C}(UV) &= x_a + x_b - \frac{x_a x_b}{C_f^{(M)}}, \\ \mathbb{E}\,\mathrm{tr}_M\, UV\mathcal{C}(UV) &= (x_a + x_b)\chi_{ab}^{(M)}, \\ C_f^{(M)} &= \frac{M(N-M)(N+1)}{N}. \end{aligned} \tag{6}$$

For completeness, the second identity has a short direct proof. The product rule gives $\mathcal{C}(UV) = (x_a + x_b)UV + \sum_{ij}[E_{ij}, U][E_{ji}, V]$. For a fixed pair $E = E_{ij}, F = E_{ji}$, the extra trace, averaged over U, is

$$\mathrm{tr}_M\{[\Phi_a(VE) - \Phi_a(V)E][F, V]\} = 0. \tag{7}$$

Indeed Φ_a is self-adjoint for the trace pairing and $\Phi_a([F, V]) = \chi_{ab}^{(M)}[F, V]$ by equivariance. The two terms cancel. The uninserted exchanged trace is $\mathbb{E}\,\mathrm{tr}\, UVUV = \chi_{ab}^{(M)}$. This is an application of the established covariance decomposition [1].

Return to $m = N/2$ and put $n = m + 1$. The removal map satisfies

$$\|\,\mathrm{Rem}\, X_r\|_{2,m}^2 = \frac{m}{n}[mn - x_r]\|X_r\|_{2,m-1}^2. \tag{8}$$

Let $C_a = \mathbb{E}h_a^2$, $\omega_a = C_a/C$, and

$$R_a = \frac{C_a^-}{C_a} = \frac{(m-p)^2 n}{m(mn - x_a)}, \qquad C_-^f = \frac{(m-1)n(N+1)}{N}. \tag{9}$$

Using (6), the norm of the removed lower product, per ordered input pair divided by $C_a C_b$, is

$$R_a R_b \frac{m}{n}\left\{(mn - x_a - x_b)(1 + \chi_{ab}^-) + \frac{x_a x_b}{C_-^f}\right\}. \tag{10}$$

Here $\chi_{ab}^- = \chi_{ab}^{(m-1)}$. Ordinary Gaussian decoupling includes the $a = b$ coefficient in the ordered-pair sum; no independent-disorder replacement of A is made.

3 Finite formula and explicit cancellation

Let $\chi = \chi_{ab}^{(m)}$ and ξ be the adjacent-charge kernel from the preceding note. Its lower parameters are $(1-m, -m, -N-1)$. If c_j is the jth summand of χ, then

$$\xi_j = \frac{m}{m-j} c_j, \qquad (\chi^-)_j = \left[1 + \frac{j}{(m+1)(m-j)}\right] c_j. \tag{11}$$

Thus $\chi^- = \chi + (\xi - \chi)/n$ exactly. Define

$$g = \frac{(m-p)^2}{m}, \quad h_0 = 2p - m, \quad \eta = \frac{p^2}{m}, \quad P = \frac{g^2 x_a x_b}{(mn - x_a)(mn - x_b)}, \quad D = \xi - \chi. \tag{12}$$

Combining (10) with the known coherent cross and the upper-product norm yields

$$\begin{aligned} \|O\|_{2,m}^2 / C^2 &= \sum_{a,b} \omega_a \omega_b f_{ab}, \\ f_{ab} &= \eta^2(1+\chi) + P\left(\frac{mn}{C_-^f} - 1 - \chi\right) + \left(\frac{g^2 - P}{n} + 2h_0 g\right) D. \end{aligned} \tag{13}$$

Every f_{ab} is a nonnegative quadratic-coefficient norm. The individual terms in this formula have large cancellations.

Set $a = p - u$, $b = p - v$, $m = p^2/(2\lambda)$ for the expansion. With $c_j^{(0)} = (-4\lambda)^j / j!$,

$$\begin{aligned} c_j &= c_j^{(0)} \left[1 + \frac{L_1}{p} + \frac{L_2 + L_1^2/2}{p^2} + \cdots\right], \\ L_1 &= -j(u + v + 2\lambda) - j(j-1), \\ L_2 &= -\frac{j}{2}(u^2 + v^2) - \frac{j(j-1)}{2}(u+v) - \frac{j(j-1)(2j-1)}{6} \\ &\quad + \lambda j(u + v + 1) + \frac{3\lambda j(j-1)}{2} - \lambda^2 j. \end{aligned} \tag{14}$$

Put $K = (g^2 - P)/n + 2h_0 g$. The coefficient multiplying c_j in (13) is $\eta^2 - P + Kj/(m-j)$, with expansion

$$\begin{aligned} p^2 r_2 + p r_1 + r_0 + O(p^{-1}), \qquad & r_2 = -\frac{j + 4\lambda}{\lambda}, \\ & r_1 = 4(2j + 2\lambda + u + v), \\ & r_0 = -2j^2 + (1 - 20\lambda)j + 4\lambda(u + v + 2) - 4uv. \end{aligned} \tag{15}$$

The unsigned part $\eta^2 + P(mn/C_-^f - 1)$ tends to $4\lambda(1+\lambda)$. The three series coefficients sum to

$$\sum_j c_j^{(0)} (r_2,\ r_1 + r_2 L_1,\ r_0 + r_1 L_1 + r_2[L_2 + L_1^2/2]) = (0, 0, 4\lambda^2 q). \tag{16}$$

These are polynomial identities: apply powers of $z\partial_z$ to e^z and set $z = -4\lambda$. Both divergent orders cancel, and the finite coefficient is independent of u, v.

4 Uniform control of the sums

The expansion on page 3 must survive multiplication by p^2. Here is a useful way to control the actual sums. Fix λ in a compact subset of $(0,\infty)$ and take $K_p = \lceil \log p \rceil$.

For all $a, b \le p$ and sufficiently large p, the exact term ratio gives $|c_j| \le A_\lambda^j / j!$. Moreover $m - j \ge m - p$. The entire coefficient $\eta^2 - P + Kj/(m-j)$ is bounded by $B_\lambda p^2(1+j)$. Therefore the absolute contribution from $j > K_p$, even before cancellation, tends to zero faster than any inverse power of p.

On $u, v, j \le K_p$, each logarithm used to expand c_j has its argument within $O(K_p/p)$ of one. Taylor's remainder bound, summed over the j factors, gives the displayed second-order expansion with a relative remainder

$$O\left(\frac{(1+u+v+j)^d}{p^3}\right) \tag{17}$$

for a fixed finite degree d. The rational coefficients admit the same kind of uniform expansion because their scaled denominators are bounded away from zero. After multiplication by p^2 and summation with the factorial envelope, the remainder is at most $O((\log p)^{d'}/p) \to 0$. Truncating the polynomial sums in (16) at K_p has a superpolynomially small error. Consequently

$$\sup_{0 \le u,v \le K_p} \left| f_{p-u,p-v} - [4\lambda + 4\lambda^2(1+q)] \right| \to 0. \tag{18}$$

Finally, the previous input-rank ratio bound gives $\omega_{p-u} \le A^u/u!$ up to a constant depending on λ. Outside this window it suffices to use the crude positive bound $f_{ab} \le B_\lambda p^4$. It follows either from (10) and Cauchy–Schwarz, or from $\| \operatorname{Rem} \| \le m$, bounded R_a, and the symmetric-product norm bound. Hence the total contribution of $u > K_p$ or $v > K_p$ vanishes. This proves (2) for the complete rank sum. Using $\lambda_N = p^2/N$ in the same compact bounds covers sequences with $\lambda_N \to \lambda$, not just exact rational sequences.

The argument also permits any fixed polynomial weight in u, v: outside the window its extra factor is at most a power of p, still dominated by the factorial tail. In particular the positive A norm is tight in its two input deficits. This result does not control the output deficit $S = a + b - r$ of A; that separate tail is still needed for the nonconstant memory filter.

5 The completed Gram and the surviving direction

Order the operators as $(B, L_N, \mathcal{S}_N, O)$, where L_N is the centered input-deficit quadratic operator from the preceding notes. Put $z = 1 + q$, $b = \theta(1 - q)$, $s = \theta^2 z + \theta(1 - q)$, and $a = \theta^2 z + 2\theta$. The old parity Gram, the preceding coherent cross, and (2) now give the full matrix

$$C^{-2} G_N \longrightarrow \begin{pmatrix} z & 0 & b & b \\ 0 & \theta z & 0 & 0 \\ b & 0 & s & s \\ b & 0 & s & a \end{pmatrix}. \tag{19}$$

Define $v = s - b^2/z = \theta(1 - q) + 4\theta^2 q/z$ and $d = a - s = \theta z$. Its determinant is $\theta^2 z^3 v > 0$. In particular the fourth direction is not lost in double scaling. The finite Gram of the three old directions has an invertible limit, so its Schur complement converges to that of (19).

For an explicit limiting orthogonalization let $c = b/z$, $Y = O - cB$, $\mathcal{S}_\perp = \mathcal{S}_N - cB$, and $\rho = v/(v + d)$. Then

$$T_N = \mathcal{S}_\perp - \rho Y \tag{20}$$

has asymptotically zero inner product with B, O, L_N, and $\|T_N\|^2 \to C^2 v d/(v + d)$, giving (4). The previously proved bridge

$$J_H F - 4\lambda q C L_N \longrightarrow 0 \quad \text{in } L^2 \tag{21}$$

can now also be paired with O, since (2) supplies a bounded fixed-sector norm. Thus the same conclusion holds for the span $(B, O, J_H F)$.

Equation (3) follows directly from $\langle \mathcal{S}_N, O \rangle \to \|\mathcal{S}_N\|^2$. It explains why the nonzero cross found in the preceding note does not eliminate the output-deficit direction: $O - \mathcal{S}_N$ retains its own positive strength. The variance of Y is now

$$V_S + \Delta_* = 4J_s^4 \left[\lambda^{-1} + \frac{4q}{1 + q} \right]. \tag{22}$$

It agrees with the earlier full-Fock expression, but here it was derived in the fixed half-filled state rather than imported from that expression.

This is linear independence in a specified equal-time span. A complete projection onto all conserved operators and a real-time spectral measure for T_N have not been obtained. A new bulk particle does not follow solely from this positive norm.

6 Evidence and next calculation

The four saved computational records support distinct parts of the derivation:

- The general-charge Casimir identity is checked using all 68,544 ordered Gaussian coordinate pairs over 15 rank-pair cases; the 45 exchanged, ordered, and uninserted identities are exact.
- The lower-charge contiguous relation has 384 finite exact checks. The raw and combined norm formulas agree in 1,200 cases. The latter also reproduce the three previous complete CAR norms 28, 81, and 2010 at covariance $w = 1$.
- Symbolic term-ratio checks verify L_1, L_2. The two divergent coefficients vanish identically in λ, u, v; the finite coefficient, Gram determinant, Schur complement, and small-λ value 16 are checked symbolically.
- Twelve all-rank calculations at $\lambda = 0.1, 0.25, 0.5$ and $p = 16, 32, 64, 128$ are evaluated at 80 and 120 digits. The finite $(B, \mathcal{X}_N/N, O)$ Gram is independently compared with all 27 entries of the three saved CAR Grams, and two further norms with exact rational pair sums.

The numerical difference norm and Schur complement use the Casimir proxy $\mathcal{X}_N/N$, not the finite $\mathcal{S}_N$. The preceding L^2 proxy bound and the now bounded O norm justify the same limiting targets. At $\lambda = 0.5, p = 128$ the three quantities (divided by C^2) differ from their limits by approximately -0.0310954, -0.0126200, and -0.0261320. These finite-p deviations are retained. Cancellation factors and precision refinement are recorded; no interval-rounding certificate is claimed.

The next missing estimate is the *output*-rank tail of the positive A norm after the adjacent-charge cancellation. Together with the input-tail result on page 4, it should permit the actual memory coefficient $g_A(a, b; r)$ to be replaced by its fixed-deficit limit in L^2. The central charge window and the low-chaos terms of the fourth-order descendants must then be restored to assemble the raw second memory coefficient. Finite-memory correlations and the bulk space/mass dictionary remain subsequent tasks.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems: Results from Embedded Unitary Ensembles*, arXiv:1504.00759, Eqs. (8)–(14), (34)–(35), and Sec. 7. The covariance and recoupling framework is used here; its dilute asymptotics are not the half-filled limit of this note.